

CPR E 4890 Final Lab Experiment (Optional)

Lab Report (Total Points: 100)

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1) Lease details screenshots (5 points)

a. eNBLease:

☐	eNBLease	2026-04-28 16:49 UTC	2026-04-28 18:49 UTC	ACTIVE	No	Delete Lease
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b. UELease:

☐	UELease	2026-04-28 16:51 UTC	2026-04-28 18:51 UTC	ACTIVE	No	Delete Lease
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2) Container overview screenshots (5 points)

a. eNBContainer:

☐	Name ^	Image	Status	Task State	
☐	> eNBContainer	arawirelesshub/srsran:srsenb	Running	-	Refresh ▾

b. UEContainer:

☐	> UEContainer	arawirelesshub/srsran:srsue_4g	Running	-	Refresh ▾
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Question: How do the specifications (e.g., CPU, memory) of the eNB and UE containers influence network performance? What security considerations should be kept in mind when configuring these resources?

Higher CPU allocation (6 cores) allows the containers to process radio signals and protocol stacks faster, which then reduces latency. More memory (8192 MB) supports larger data buffers which prevents packet drops under heavy load. Security considerations would include things like using strong passwords for root access, not exposing container ports to the public internet, or possibly avoiding default credentials since these containers run real radio software that could be exploited if left open.

3) Starting the EPC and eNB - Analysis Questions (10 points)

```

root@eNB:~# srsepc

Built in RelWithDebInfo mode using 20.10.1.

--- Software Radio Systems EPC ---

Reading configuration file /root/.config/srslte/epc.conf...
HSS Initialized.
MME S11 Initialized
MME GTP-C Initialized
MME Initialized. MCC: 0xf001, MNC: 0xff01
SPGW GTP-U Initialized.
SPGW S11 Initialized.
SP-GW Initialized.
Received S1 Setup Request.
S1 Setup Request - eNB Name: srsenb01, eNB id: 0x19b
S1 Setup Request - MCC:001, MNC:01
S1 Setup Request - TAC 7, B-PLMN 0xf110
S1 Setup Request - Paging DRX v128
Sending S1 Setup Response
Initial UE message: LIBLTE_MME_MSG_TYPE_ATTACH_REQUEST
Received Initial UE message -- Attach Request
Attach request -- M-TMSI: 0x604ae27c
Attach request -- eNB-UE S1AP Id: 1
Attach request -- Attach type: 1
Attach Request -- UE Network Capabilities EEA: 11110000
Attach Request -- UE Network Capabilities EIA: 01110000
Attach Request -- MS Network Capabilities Present: false
PDN Connectivity Request -- EPS Bearer Identity requested: 0
PDN Connectivity Request -- Procedure Transaction Id: 1
PDN Connectivity Request -- ESM Information Transfer requested: false
UL NAS: Received Identity Response
ID Response -- IMSI: 001010123456789
Downlink NAS: Sent Authentication Request
UL NAS: Received Authentication Response
Authentication Response -- IMSI 001010123456789
UE Authentication Accepted.
Generating KeNB with UL NAS COUNT: 0
Downlink NAS: Sending NAS Security Mode Command.
UL NAS: Received Security Mode Complete
Security Mode Command Complete -- IMSI: 001010123456789
Getting subscription information -- QCI 7
Sending Create Session Request.
Creating Session Response -- IMSI: 1010123456789
Creating Session Response -- MME control TEID: 1
Received GTP-C PDU. Message type: GTPC_MSG_TYPE_CREATE_SESSION_REQUEST

SPGW: Allocated Ctrl TEID 1
SPGW: Allocated User TEID 1
SPGW: Allocate UE IP 172.16.0.2
Received Create Session Response
Create Session Response -- SPGW control TEID 1
Create Session Response -- SPGW S1-U Address: 127.0.1.100
SPGW Allocated IP 172.16.0.2 to IMSI 001010123456789
Adding attach accept to Initial Context Setup Request
Sent Initial Context Setup Request. E-RAB id 5
Received Initial Context Setup Response
E-RAB Context Setup. E-RAB id 5
E-RAB Context -- eNB TEID 0x1; eNB GTP-U Address 127.0.1.1
UL NAS: Received Attach Complete
Unpacked Attached Complete Message. IMSI 1010123456789
Unpacked Activate Default EPS Bearer message. EPS Bearer id 5
Received GTP-C PDU. Message type: GTPC_MSG_TYPE_MODIFY_BEARER_REQUEST
Sending EMM Information

```

```

root@eNB:~# srsenb
--- Software Radio Systems LTE eNodeB ---

Reading configuration file /root/.config/srslte/enb.conf...
WARNING: cpu0 scaling governor is not set to performance mode. Realtime processing could be compromised. Consider setting it to performance mode before running the application.

Built in RelWithDebInfo mode using 20.10.1.

Opening 1 channels in RF device=uhd with args=default
[INFO] [UHD] linux; GNU C++ version 9.4.0; Boost_107100; UHD_3.15.0-HEAD-0-gaea0e2de
[INFO] [LOGGING] Fastpath logging disabled at runtime.
Opening USRP channels=1, args: type=b200,master_clock_rate=23.04e6
[INFO] [B200] Detected Device: B210
[INFO] [B200] Loading FPGA image: /usr/local/share/uhd/images/usrp_b210_fpga.bin...
[INFO] [B200] Operating over USB 3.
[INFO] [B200] Detecting internal GPSDO....
[INFO] [GPS] No GPSDO found
[INFO] [B200] Initialize CODEC control...
[INFO] [B200] Initialize Radio control...
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Asking for clock rate 23.040000 MHz...
[INFO] [B200] Actually got clock rate 23.040000 MHz.
Setting frequency: DL=2680.0 Mhz, UL=2560.0 Mhz for cc_idx=0

==== eNodeB started ===
Type <t> to view trace
RACH: tti=6901, cc=0, preamble=0, offset=1, temp_crnti=0x46
RACH: tti=6901, cc=0, preamble=3, offset=7, temp_crnti=0x47
RACH: tti=6901, cc=0, preamble=30, offset=31, temp_crnti=0x48
Disconnecting rnti=0x47.
Disconnecting rnti=0x48.
User 0x46 connected

```

After starting **srsepc** and **srsenb**, answer:

a. What three main components are initialized?

HSS (Home Subscriber Server), MME (Mobility Management Entity), and SPGW (Serving and PDN Gateway)

b. What MCC and MNC values do you see?

MCC: 001, MNC: 01 (seen in the EPC terminal as “S1 Setup Request - MCC:001, MNC:01”)

c. What does “SPGW” stand for and what is its role?

SPGW stands for Serving and PDN Gateway. It handles data routing between the UE and external networks, allocates IP addresses to connected devices, and manages data tunnels between the radio network and the core.

d. What did you observe when srsenb connected to the EPC?

When srsenb started, it detected the USRP B210 SDR hardware and initialized the radio at DL=2680.0 MHz and UL=2560.0 MHz. It then sent an S1 Setup Request to the EPC. The EPC responded with an S1 Setup Response, confirming the connection. Once the UE attached, RACH messages appeared (preamble attempts at tti=6901), and “User 0x46 connected” confirmed a successful UE attachment.

```

root@UE:~# srsue
Reading configuration file /root/.config/srslte/ue.conf...
WARNING: cpu0 scaling governor is not set to performance mode. Realtime processing could be compromised. Consider setting it to performance mode before running the application.

Built in RelWithDebInfo mode using 20.10.1.

Opening 1 channels in RF device=default with args=default
[INFO] [UHD] linux; GNU C++ version 9.4.0; Boost 107100; UHD_3.15.0-HEAD-0-gaea0e2de
[INFO] [LOGGING] Fastpath logging disabled at runtime.
Opening USRP channels=1, args: type=b200,master_clock_rate=23.04e6
[INFO] [B200] Detected Device: B210
[INFO] [B200] Loading FPGA image: /usr/local/share/uhd/images/usrp_b210_fpga.bin...
[INFO] [B200] Operating over USB 3.
[INFO] [B200] Detecting internal GPSDO...
[INFO] [GPS] No GPSDO found
[INFO] [B200] Initialize CODEC control...
[INFO] [B200] Initialize Radio control...
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Asking for clock rate 23.040000 MHz...
[INFO] [B200] Actually got clock rate 23.040000 MHz.
Waiting PHY to initialize ... done!
Attaching UE...
.
Found Cell: Mode=FDD, PCI=1, PRB=50, Ports=1, CF0=-0.3 KHz
Found PLMN: Id=00101, TAC=7
Random Access Transmission: seq=0, ra-rnti=0x2
Random Access Complete. c-rnti=0x46, ta=1
RRC Connected
Network attach successful. IP: 172.16.0.2
Software Radio Systems LTE (srslte) 20/4/2026 17:1:24 TZ:0

```

4) Starting the UE - Connection Process (10 points)

After starting srsue, record:

a. Cell found - What's the PCI and PRB count?

PCI = 1, PRB = 50

b. PLMN found - Does it match the network configured in EPC?

Yes. The UE found PLMN Id=00101 (MCC=001, MNC=01), which matches the MCC:001, MNC:01 configured in the EPC.

c. What's your C-RNTI?

0x46

d. What IP address was assigned?

172.16.0.2

From the EPC terminal, record:

- IMSI number: 001010123456789
- Allocated IP: 172.16.0.2
- Bearer QCI: 7

Screenshot of ping 172.16.0.1 results:

```

dhiman_iastate@ara-jbox:~$ ssh root@10.189.10.221
root@10.189.10.221's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.19.17-051917-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Tue Apr 28 11:59:16 2026 from 10.188.0.15
root@UE:~# ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1) 56(84) bytes of data.
64 bytes from 172.16.0.1: icmp_seq=1 ttl=64 time=18.0 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=64 time=36.8 ms
64 bytes from 172.16.0.1: icmp_seq=3 ttl=64 time=34.8 ms
64 bytes from 172.16.0.1: icmp_seq=4 ttl=64 time=32.8 ms
64 bytes from 172.16.0.1: icmp_seq=5 ttl=64 time=30.8 ms
^C
--- 172.16.0.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 17.978/30.633/36.776/6.632 ms
root@UE:~#

```

Question: Why is it crucial to start the EPC before the eNB? How does the interaction between these components affect the overall network functionality?

The EPC needs to start first so that the eNB can register with the EPC using the S1 interface before it can serve any of the UEs. If the eNB starts without a running EPC, it basically has nowhere to send the S1 Setup Request and will fail. The EPC handles auth, IP allocation, and session management and without it the eNB can't authenticate users or route the data.

5) Console trace screenshots (10 points)

a. eNBContainer:

```

root@eNB: ~ -- ssh -i ~/ssh/ARA dhiman_iasta...
root@eNB: ~ -- ssh -i ~/ssh/ARA dhiman_iasta...

t
Enter t to stop trace.

-----DL-----UL-----
rntti cqi ri mcs brate ok nok (%) snr phr mcs brate ok nok (%) bsr
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 8.9 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 8.8 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0

-----DL-----UL-----
rntti cqi ri mcs brate ok nok (%) snr phr mcs brate ok nok (%) bsr
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 8.8 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0

-----DL-----UL-----
rntti cqi ri mcs brate ok nok (%) snr phr mcs brate ok nok (%) bsr
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 8.7 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0
46 9.0 0 0.0 0 0 0 0% 0 0.0 0 0 0 0 0% 0.0

```


b. UEContainer:

```
t
Enter t to stop trace.
-----Signal-----DL-----UL-----
cc pci rsrp pl cfo mcs snr turbo brate bler ta_us mcs buff brate bler
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -349 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -349 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -349 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -349 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -349 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -349 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
-----Signal-----DL-----UL-----
cc pci rsrp pl cfo mcs snr turbo brate bler ta_us mcs buff brate bler
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -349 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -349 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 20 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
-----Signal-----DL-----UL-----
cc pci rsrp pl cfo mcs snr turbo brate bler ta_us mcs buff brate bler
0 1 -79 79 -351 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 20 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -349 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -349 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
0 1 -79 79 -350 0.0 19 0.0 0.0 0% 0.52 0.0 0.0 0.0 0%
-----Signal-----DL-----UL-----
```

Analysis: Analyze and discuss what you see in the console trace for the 2 containers.

The eNB console trace shows rnti=46, cqi=9 for the connected UE. The DL brate and UL brate both read 0.0 Mbps because no iperf traffic is flowing yet in this snapshot. The trace confirms the UE is attached and the eNB is actively tracking it.

The UE console trace shows cc=0, pci=1, rsrp=-79 dBm, path loss=79 dB, cfo=-350 Hz, and SNR=19–20 dB. These values indicate a stable connection with good signal quality. The brate is 0.0 here as well since iperf had not started. The ta_us value of 0.52 shows a small timing advance, meaning the UE is close to the eNB with low propagation delay.

6) Define the options used by the UEContainer iperf command (5 points)

For the command: **iperf -c 172.16.0.1 -u -i 1 -b 25M -t 10**

- **-c:** This flag runs iperf in client mode and connects to the server at IP 172.16.0.1
- **-u:** This flag forces the use of UDP instead of TCP
- **-i:** This flag sets the reporting interval to 1 second
- **-b:** This flag sets the target sending bandwidth to 25 Mbits/sec
- **-t:** This flag sets the test duration to 10 seconds

7) Define the brate metric seen on the console trace (5 points)

Definition of brate: brate (bit rate) is the actual throughput measured in Mbps and it indicates how many bits per second were successfully transmitted and received during the test interval.

8) Variable bandwidth testing (10 points)

Test at least 3 different -b values and record:

- Test 1 - b value: 20M Achieved throughput: 16.2 Mbits/Sec

```
root@UE:~# iperf -c 172.16.0.1 -u -i 1 -b 20M -t 10
-----
Client connecting to 172.16.0.1, UDP port 5001
Sending 1470 byte datagrams, IPG target: 560.76 us (kalman adjust)
UDP buffer size: 208 KByte (default)
-----
[ 3] local 172.16.0.2 port 36723 connected with 172.16.0.1 port 5001
[ ID] Interval      Transfer    Bandwidth
[ 3] 0.0- 1.0 sec   2.50 MBytes 21.0 Mbits/sec
[ 3] 1.0- 2.0 sec   2.50 MBytes 21.0 Mbits/sec
[ 3] 2.0- 3.0 sec   2.50 MBytes 21.0 Mbits/sec
[ 3] 3.0- 4.0 sec   2.50 MBytes 21.0 Mbits/sec
[ 3] 4.0- 5.0 sec   2.50 MBytes 21.0 Mbits/sec
[ 3] 5.0- 6.0 sec   2.50 MBytes 21.0 Mbits/sec
[ 3] 6.0- 7.0 sec   2.50 MBytes 21.0 Mbits/sec
[ 3] 7.0- 8.0 sec   2.50 MBytes 21.0 Mbits/sec
[ 3] 8.0- 9.0 sec   2.50 MBytes 21.0 Mbits/sec
[ 3] 9.0-10.0 sec   25.0 MBytes 21.0 Mbits/sec
[ 3] Sent 17833 datagrams
[ 3] Server Report:
[ 3] 0.0-10.2 sec  19.6 MBytes 16.2 Mbits/sec  0.427 ms 3849/17833 (22%)
root@UE:~#
```

- Test 2 - b value: 25M Achieved throughput: 15.7 Mbits/Sec

```
root@UE:~# iperf -c 172.16.0.1 -u -i 1 -b 25M -t 10
-----
Client connecting to 172.16.0.1, UDP port 5001
Sending 1470 byte datagrams, IPG target: 448.61 us (kalman adjust)
UDP buffer size: 208 KByte (default)
-----
[ 3] local 172.16.0.2 port 52283 connected with 172.16.0.1 port 5001
[ ID] Interval      Transfer    Bandwidth
[ 3] 0.0- 1.0 sec   3.13 MBytes 26.2 Mbits/sec
[ 3] 1.0- 2.0 sec   3.12 MBytes 26.2 Mbits/sec
[ 3] 2.0- 3.0 sec   3.12 MBytes 26.2 Mbits/sec
[ 3] 3.0- 4.0 sec   3.12 MBytes 26.2 Mbits/sec
[ 3] 4.0- 5.0 sec   3.12 MBytes 26.2 Mbits/sec
[ 3] 5.0- 6.0 sec   3.12 MBytes 26.2 Mbits/sec
[ 3] 6.0- 7.0 sec   3.12 MBytes 26.2 Mbits/sec
[ 3] 7.0- 8.0 sec   3.12 MBytes 26.2 Mbits/sec
[ 3] 8.0- 9.0 sec   3.12 MBytes 26.2 Mbits/sec
[ 3] 9.0-10.0 sec   31.3 MBytes 26.2 Mbits/sec
[ 3] Sent 22292 datagrams
[ 3] Server Report:
[ 3] 0.0-10.2 sec  19.1 MBytes 15.7 Mbits/sec  4.538 ms 8665/22293 (39%)
root@UE:~#
```

- Test 3 - b value: 20M Achieved throughput: 15.9 Mbits/Sec

```

root@UE:~# iperf -c 172.16.0.1 -u -i 1 -b 30M -t 10
-----
Client connecting to 172.16.0.1, UDP port 5001
Sending 1470 byte datagrams, IPG target: 373.84 us (kalman adjust)
UDP buffer size: 208 KByte (default)
-----
[ 3] local 172.16.0.2 port 43701 connected with 172.16.0.1 port 5001
[ ID] Interval      Transfer      Bandwidth
[ 3] 0.0- 1.0 sec    3.75 MBytes   31.5 Mbits/sec
[ 3] 1.0- 2.0 sec    3.75 MBytes   31.5 Mbits/sec
[ 3] 2.0- 3.0 sec    3.75 MBytes   31.5 Mbits/sec
[ 3] 3.0- 4.0 sec    3.75 MBytes   31.5 Mbits/sec
[ 3] 4.0- 5.0 sec    3.75 MBytes   31.5 Mbits/sec
[ 3] 5.0- 6.0 sec    3.75 MBytes   31.5 Mbits/sec
[ 3] 6.0- 7.0 sec    3.75 MBytes   31.5 Mbits/sec
[ 3] 7.0- 8.0 sec    3.75 MBytes   31.5 Mbits/sec
[ 3] 8.0- 9.0 sec    3.75 MBytes   31.5 Mbits/sec
[ 3] 9.0-10.0 sec    3.75 MBytes   31.5 Mbits/sec
[ 3] 0.0-10.0 sec    37.5 MBytes   31.5 Mbits/sec
[ 3] Sent 26750 datagrams
[ 3] Server Report:
[ 3] 0.0-10.3 sec   19.5 MBytes   15.9 Mbits/sec    6.028 ms 12875/26751 (48%)
root@UE:~#

```

Analysis:

- **At what point does the requested bandwidth exceed the achievable throughput?**

The network starts saturating at 25M. At 20M the packet loss is 22%, but once the target is raised to 25M, the loss jumps to 39% while throughput actually drops slightly to 15.7 Mbps. This shows the link maximum is around 16 Mbps.

- **What indicates saturation?**

Saturation is indicated by the packet loss steadily increasing (22% to 39% to 48%) while the server-side throughput stays flat around 15.7–16.2 Mbps regardless of how high the target bandwidth is set.

- **How does adjusting the bandwidth allocation affect throughput and latency?**

Below the saturation point (roughly 20M), throughput scales with the requested bandwidth. Above it, the throughput stops improving but packet loss increases and the jitter gets worse, which increases the effective latency. The network cannot deliver more than approximately 16 Mbps on this link.

9) TCP Performance Testing (10 points)

TCP Throughput Results: 16.9 Mbps

```

root@UE:~# iperf -c 172.16.0.1 -t 10 -i 1
-----
Client connecting to 172.16.0.1, TCP port 5001
TCP window size: 85.0 KByte (default)
-----
[ 3] local 172.16.0.2 port 47736 connected with 172.16.0.1 port 5001
[ ID] Interval      Transfer      Bandwidth
[ 3] 0.0- 1.0 sec    2.38 MBytes   19.9 Mbits/sec
[ 3] 1.0- 2.0 sec    2.25 MBytes   18.9 Mbits/sec
[ 3] 2.0- 3.0 sec    2.12 MBytes   17.8 Mbits/sec
[ 3] 3.0- 4.0 sec    2.25 MBytes   18.9 Mbits/sec
[ 3] 4.0- 5.0 sec    2.50 MBytes   21.0 Mbits/sec
[ 3] 5.0- 6.0 sec    1.62 MBytes   13.6 Mbits/sec
[ 3] 6.0- 7.0 sec    1.51 MBytes   12.6 Mbits/sec
[ 3] 7.0- 8.0 sec    2.38 MBytes   19.9 Mbits/sec
[ 3] 8.0- 9.0 sec    1.62 MBytes   13.6 Mbits/sec
[ 3] 9.0-10.0 sec    1.62 MBytes   13.6 Mbits/sec
[ 3] 0.0-10.1 sec    20.3 MBytes   16.9 Mbits/sec
root@UE:~#

```

```

^Croot@eNB:~# iperf -s
-----
Server listening on TCP port 5001
TCP window size: 128 KByte (default)
-----
[ 4] local 172.16.0.1 port 5001 connected with 172.16.0.2 port 47736
[ ID] Interval      Transfer      Bandwidth
[ 4] 0.0-10.8 sec    20.3 MBytes   15.7 Mbits/sec

```

Comparison:

- Document the TCP throughput and compare with your UDP results. Why might they differ?

TCP achieved about 16.9 Mbps, which is very similar to UDP. However, TCP handles errors through retransmissions rather than just dropping packets. This means TCP throughput is more consistent but can vary more over time as it adjusts its congestion window. UDP at 25M showed 39% packet loss, while TCP automatically slows down before that happens. The slightly lower TCP throughput compared to UDP's client-side rate is expected because TCP has additional overhead from acknowledgment packets and connection management.

10) Bidirectional Testing (10 points)

```
root@UE:~# iperf -c 172.16.0.1 -u -b 20M -d -t 10 -i 1
-----
Client connecting to 172.16.0.1, UDP port 5001
Sending 1470 byte datagrams, IPG target: 560.76 us (kalman adjust)
UDP buffer size: 208 KByte (default)
-----
Server listening on UDP port 5001
Receiving 1470 byte datagrams
UDP buffer size: 208 KByte (default)
-----
[ 3] local 172.16.0.2 port 48435 connected with 172.16.0.1 port 5001 (peer 2.0.13)
[ 5] local 172.16.0.2 port 5001 connected with 172.16.0.1 port 35104
[10] Interval      Transfer      Bandwidth
[ 3] 0.0- 1.0 sec  2.43 MBytes  20.4 Mbits/sec
[ 5] 0.0- 1.0 sec  2.16 MBytes  18.2 Mbits/sec    0.476 ms   0/ 1544 (0%)
[ 3] 1.0- 2.0 sec  2.50 MBytes  21.0 Mbits/sec
[ 5] 1.0- 2.0 sec  2.27 MBytes  19.0 Mbits/sec    0.453 ms 128/ 1744 (7.3%)
[ 3] 2.0- 3.0 sec  2.50 MBytes  21.0 Mbits/sec
[ 5] 2.0- 3.0 sec  2.24 MBytes  18.8 Mbits/sec    0.485 ms 183/ 1782 (10%)
[ 3] 3.0- 4.0 sec  2.50 MBytes  21.0 Mbits/sec
[ 5] 3.0- 4.0 sec  2.28 MBytes  19.1 Mbits/sec    0.436 ms 158/ 1784 (8.9%)
[ 3] 4.0- 5.0 sec  2.50 MBytes  21.0 Mbits/sec
[ 5] 4.0- 5.0 sec  2.28 MBytes  19.1 Mbits/sec    0.462 ms 159/ 1783 (8.9%)
[ 3] 5.0- 6.0 sec  2.50 MBytes  21.0 Mbits/sec
[ 5] 5.0- 6.0 sec  2.23 MBytes  18.7 Mbits/sec    0.430 ms 191/ 1785 (11%)
[ 3] 6.0- 7.0 sec  2.50 MBytes  21.0 Mbits/sec
[ 5] 6.0- 7.0 sec  2.24 MBytes  18.8 Mbits/sec    0.495 ms 159/ 1754 (9.1%)
[ 3] 7.0- 8.0 sec  2.50 MBytes  21.0 Mbits/sec
[ 5] 7.0- 8.0 sec  2.18 MBytes  18.3 Mbits/sec    0.434 ms 257/ 1811 (14%)
[ 3] 8.0- 9.0 sec  2.50 MBytes  21.0 Mbits/sec
[ 5] 8.0- 9.0 sec  2.28 MBytes  19.1 Mbits/sec    0.443 ms 160/ 1785 (9%)
[ 3] 9.0-10.0 sec  2.50 MBytes  21.0 Mbits/sec
[ 5] 9.0-10.0 sec  2.28 MBytes  19.1 Mbits/sec    0.478 ms 158/ 1782 (8.9%)
[ 5] 0.0-10.0 sec  22.4 MBytes  18.8 Mbits/sec    0.478 ms 1553/17554 (8.8%)
[SUM] 0.0-10.0 sec  24.6 MBytes  20.6 Mbits/sec    0.478 ms 1553/19098 (8.1%)
[ 5] WARNING: ack of last datagram failed after 10 tries.
[ 3] 0.0-10.0 sec  25.0 MBytes  20.9 Mbits/sec
[ 3] Sent 17834 datagrams
[ 3] Server Report:
[ 3] 0.0-10.2 sec  19.1 MBytes  15.7 Mbits/sec    0.428 ms 4191/17834 (24%)
[ 3] 0.0000-10.1989 sec  1 datagrams received out-of-order
root@UE:~#
```

```
^Croot@eNB:~# iperf -s -u
-----
Server listening on UDP port 5001
Receiving 1470 byte datagrams
UDP buffer size: 208 KByte (default)
-----
[ 3] local 172.16.0.1 port 5001 connected with 172.16.0.2 port 48435 (peer 2.0.13)
-----
Client connecting to 172.16.0.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 560.76 us (kalman adjust)
UDP buffer size: 208 KByte (default)
-----
[ 5] local 172.16.0.1 port 35104 connected with 172.16.0.2 port 5001
[10] Interval      Transfer      Bandwidth      Jitter  Lost/Total Datagrams
[ 3] 0.0-10.2 sec  19.1 MBytes  15.7 Mbits/sec    0.429 ms 4191/17834 (24%)
[ 3] 0.0000-10.1989 sec  1 datagrams received out-of-order
[ 5] 0.0-10.0 sec  25.0 MBytes  21.0 Mbits/sec
[ 5] Sent 17834 datagrams
[ 5] Server Report:
[ 5] 0.0-10.0 sec  22.4 MBytes  18.8 Mbits/sec    0.477 ms 1553/17554 (8.8%)
```

Uplink throughput: 15.7 Mbps

Downlink throughput: 18.8 Mbps

Total combined: 34.5 Mbps

Analysis:

- How does simultaneous traffic affect performance?

Running both directions at the same time reduces the available resources for each individual direction. The uplink dropped to 15.7 Mbps (with 24% packet loss) compared to around 16 Mbps unidirectional, and the downlink was 18.8 Mbps. Each direction had more packet loss than in the unidirectional tests.

- **Is the total bidirectional throughput equal to the sum of unidirectional tests? Explain.**

No, not exactly. The unidirectional UDP test at 20M target gave about 16.2 Mbps in one direction. If the link were perfectly symmetrical and fully separated, bidirectional would be $2 \times 16.2 = 32.4$ Mbps. The actual total was roughly 34.5 Mbps, which is slightly higher because LTE uses FDD (Frequency Division Duplex) where the uplink and downlink operate on separate frequencies simultaneously. However, both directions still share the same hardware and processing resources, so there is some interference and each direction performs slightly worse than in isolation.

11) Distance Impact on Throughput (10 points)

Original location:

- Host number: 009
- Throughput: 15.7 Mbps
- Signal metrics: RSRP: -79 dBm SNR: 19-20 dB CFO: -350 Hz

New location (farther):

- Host number: 020
- Throughput: 20.4 Mbps
- Signal metrics: RSRP: _____ SNR: _____ CFO: _____ (DID NOT RUN)

```

root@UE:~# iperf -c 172.16.0.1 -u -b 20M -d -t 10 -i 1
-----
Server listening on UDP port 5001
Receiving 1470 byte datagrams
UDP buffer size: 208 KByte (default)
-----

Client connecting to 172.16.0.1, UDP port 5001
Sending 1470 byte datagrams, IPG target: 560.76 us (kalman adjust)
UDP buffer size: 208 KByte (default)
-----

recvack failed: Resource temporarily unavailable
[ 4] local 10.0.4.109 port 52783 connected with 172.16.0.1 port 5001 (server version is old)
[ ID] Interval      Transfer      Bandwidth
[ 4] 0.0- 1.0 sec   1.83 MBytes   15.4 Mbits/sec
[ 4] 1.0- 2.0 sec   2.50 MBytes   21.0 Mbits/sec
[ 4] 2.0- 3.0 sec   2.50 MBytes   21.0 Mbits/sec
[ 4] 3.0- 4.0 sec   2.50 MBytes   21.0 Mbits/sec
[ 4] 4.0- 5.0 sec   2.50 MBytes   21.0 Mbits/sec
[ 4] 5.0- 6.0 sec   2.50 MBytes   21.0 Mbits/sec
[ 4] 6.0- 7.0 sec   2.50 MBytes   21.0 Mbits/sec
[ 4] 7.0- 8.0 sec   2.50 MBytes   21.0 Mbits/sec
[ 4] 8.0- 9.0 sec   2.50 MBytes   21.0 Mbits/sec
[ 4] 9.0-10.0 sec   2.50 MBytes   21.0 Mbits/sec
[ 4] WARNING: did not receive ack of last datagram after 10 tries.
[ 4] 0.0-10.3 sec  25.0 MBytes   20.4 Mbits/sec
[ 4] Sent 17834 datagrams

```

Analysis:

- **Discuss the effect distance has on throughput in a 4G network.**

In a 4G network, as the UE moves farther from the eNB, the received signal power (RSRP) decreases and the signal-to-noise ratio (SNR) worsens. This causes the eNB to assign lower-order modulation schemes (lower MCS values), which reduces the data rate. Path loss increases with distance, meaning more signal energy is lost before reaching the receiver.

- **Describe the changes you observed based on change in throughput.**

Host 009 was at a moderate distance and showed stable performance around 15.7 Mbps with RSRP=-79 dBm and SNR=19-20 dB. For host 020, the iperf client-side showed roughly 20.4 Mbps but a datagram acknowledgment warning appeared, suggesting some link instability. Without a full console trace for host 020, exact signal metrics could not be compared directly.

- **Comment on the link quality based on metrics such as RSRP, RSRQ, SNR, CFO if visible.**

For host 009, RSRP of -79 dBm is a moderate signal level (acceptable for LTE; strong is above -80 dBm). SNR of 19-20 dB is a healthy value supporting higher-order modulation. CFO of -350 Hz is a small carrier frequency offset that the system corrects automatically. If a farther host showed lower RSRP and SNR, you would expect higher BLER (block error rate) and lower brate in the console trace.

12) Empty lease page screenshot (5 points)

ARA

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Project

Project / Reservations /

Jupyter Notebook

Summary

Container

Network

Orchestration

Object Store

Reservations

Leases

Identity

Data

Leases

Host Calendar

create Lease

Lease name	Start date	End date	Status	Degraded	Actions
No items to display.					

13) Summary and Conclusions (5 points)

Provide your summary addressing:

- What was the maximum throughput achieved?
- What factors limited the performance?
- Key findings from your experiments
- Challenges encountered during the lab
- Real-world applications of your findings
- Suggested improvements for the network

The maximum throughput achieved was approximately 16.2 Mbps (UDP, unidirectional, at 20M target). The main factors limiting performance were the physical link quality (RSRP=-79 dBm, SNR=19-20 dB) and the LTE radio channel capacity, which saturated around 16 Mbps regardless of the requested bandwidth.

Key findings: the network handled UDP and TCP at similar throughput levels (roughly 16 Mbps). Increasing the UDP target bandwidth beyond saturation only increased packet loss — it did not improve throughput. Bidirectional testing showed LTE's FDD capability but each direction experienced slightly worse performance than unidirectional.

Challenges included finding host pairs with reliable RF coverage, managing multiple terminals simultaneously, and ensuring the EPC was fully initialized before starting the eNB and UE.

Real-world applications include understanding LTE capacity planning where the operators must size their networks to avoid saturation, especially for video streaming or file transfers. The findings also highlight how distance and signal quality directly affect user experience in cellular networks.

Suggested improvements: using hosts with stronger RF connectivity, increasing container CPU allocation for better real-time signal processing, and running longer iperf tests for more stable throughput averages.